

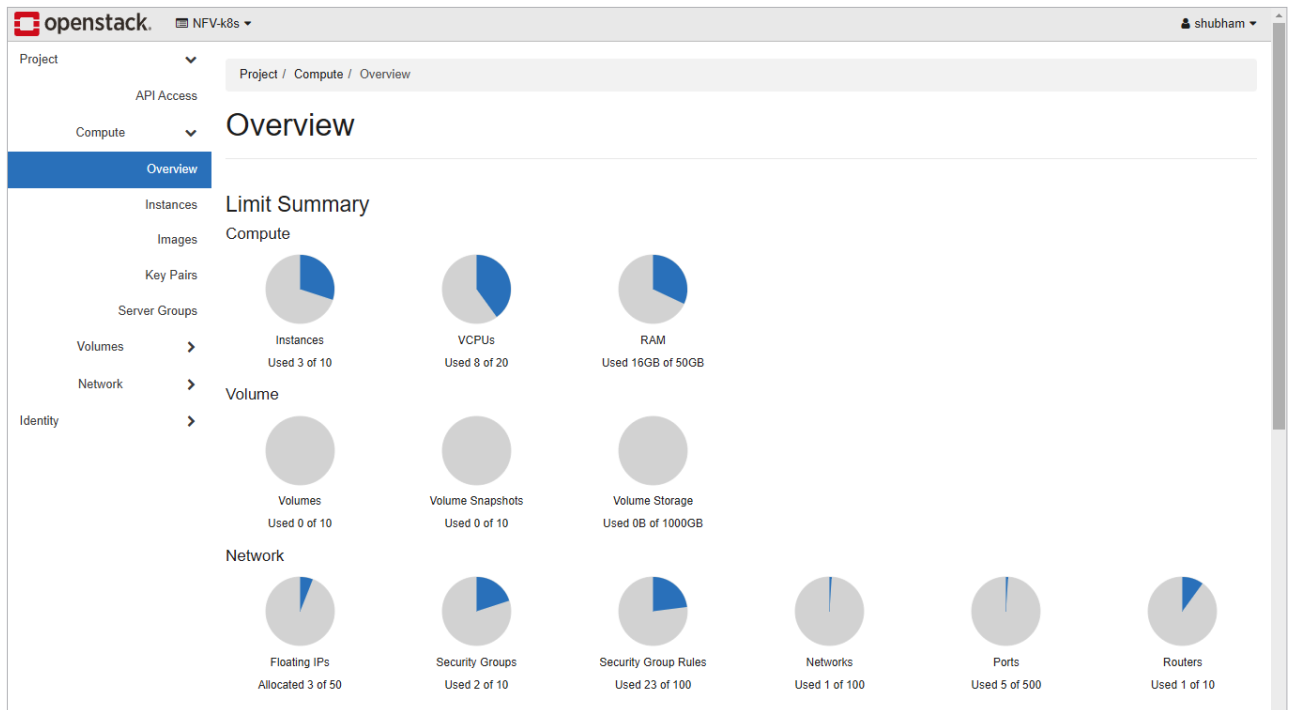


Figure 5: OpenStack dashboard overview after VM deployment

```
shubham@k8s-docker-master:~$ kubectl get all
NAME                                READY   STATUS    RESTARTS   AGE
pod/server-deployment-57c4cc49c-897c8   1/1     Running   0           58s

NAME                                TYPE               CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
service/kubernetes                  ClusterIP          10.96.0.1    <none>         443/TCP           5d2h
service/nfv-service                 LoadBalancer      10.106.43.46 192.168.122.31 80:32252/TCP     60m
service/server-cluster-ip-service    ClusterIP          10.110.218.58 <none>         80/TCP           58s

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/server-deployment    1/1     1             1           58s

NAME                                DESIRED   CURRENT   READY   AGE
replicaset.apps/server-deployment-57c4cc49c 1         1         1       58s
shubham@k8s-docker-master:~$
```

Figure 6: Kubernetes master node depicting all pods, services, and deployments after NFV deployment

the application, along with the port numbers on which the application would be running. Every running service is allocated an IP address by Kubernetes to allow communication between them, and this can be seen in

the figure. The command, as shown in the figure, prints all the various objects deployed over the Kubernetes cluster while showcasing helpful details such as status, number of restarts of the pods, and the services.

The administrator can upscale and downscale the number of deployments and replica sets based on the traffic requirements.

Figure 7 depicts all the running pods in the Kubernetes cluster. These pods are attached to different namespaces and the command asks Kubernetes to print all the pods associated with different namespaces. These pods could be running on different nodes of the cluster based on their roles. The application-based objects always run on the worker nodes and a similar one can be seen here; our server deployment can be seen running on the worker node under the NODE section.

```
shubham@k8s-docker-master:~$ kubectl get pods --all-namespaces -o wide
NAMESPACE NAME                                READY   STATUS    RESTARTS   AGE   IP             NODE               NOMINATED NODE   READINESS GATES
default   server-deployment-57c4cc49c-897c8   1/1     Running   0           85s   10.244.1.14    k8s-docker-worker <none>           <none>
kube-system coredns-64897985d-6v6ph             1/1     Running   0           5d2h   10.244.0.3     k8s-docker-master <none>           <none>
kube-system coredns-64897985d-bgrl2             1/1     Running   0           5d2h   10.244.0.2     k8s-docker-master <none>           <none>
kube-system etcd-k8s-docker-master      1/1     Running   0           5d2h   192.168.122.31 k8s-docker-master <none>           <none>
kube-system kube-apiserver-k8s-docker-master 1/1     Running   0           5d2h   192.168.122.31 k8s-docker-master <none>           <none>
kube-system kube-controller-manager-k8s-docker-master 1/1     Running   0           5d2h   192.168.122.31 k8s-docker-master <none>           <none>
kube-system kube-flannel-ds-xtcqd         1/1     Running   0           5d2h   192.168.122.32 k8s-docker-worker <none>           <none>
kube-system kube-flannel-ds-zxw4          1/1     Running   0           5d2h   192.168.122.31 k8s-docker-master <none>           <none>
kube-system kube-proxy-4cph              1/1     Running   0           5d2h   192.168.122.32 k8s-docker-worker <none>           <none>
kube-system kube-proxy-ggsgt             1/1     Running   0           5d2h   192.168.122.31 k8s-docker-master <none>           <none>
kube-system kube-scheduler-k8s-docker-master 1/1     Running   0           5d2h   192.168.122.31 k8s-docker-master <none>           <none>
kube-system metrics-server-64d96c8bf6-f8xwn 1/1     Running   0           5d2h   10.244.1.3     k8s-docker-worker <none>           <none>
shubham@k8s-docker-master:~$
```

Figure 7: Kubernetes master node depicting all pods in default namespaces after NFV deployment